EXERCISE 8

Bernoulli · Substitution z = y^(1-α)

General Bernoulli Equation Reduction

Original statement · ProblemsCh2.pdf, p. 2

PROBLEM STATEMENT

Show that the Bernoulli equation

$$y'(x) = a(x)\,y(x) + b(x)\,y(x)^{\alpha},$$

where $\alpha \neq 0, 1$, can be transformed into a linear ODE with the change of variable $z(x) = y(x)^{1-\alpha}$ and find the solution. What happens if $\alpha = 0$ or $\alpha = 1$?

Source: ProblemsCh2.pdf, Exercise 8 (page 2).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

Assume a and b are continuous on an interval I and $\alpha \in \mathbb{R}, \alpha \neq 0, 1$. The power y^{α} and $y^{1-\alpha}$ must make sense, so we work with $y > 0$ (for rational α with odd denominators other branches can be treated analogously). On $y > 0$, $f(x, y) = ay + by^{\alpha}$ and $\partial f/\partial y = a + \alpha b\, y^{\alpha-1}$ are continuous, so IVPs with $y(x_0) = y_0 > 0$ have unique local solutions.

The equation is non-linear because y^{α} with $\alpha \neq 0, 1$ is not affine in y. When $\alpha > 0$, $y \equiv 0$ is also a solution; it is excluded by the assumption $y > 0$ and has to be added by hand (for $0 < \alpha < 1$ uniqueness at $y = 0$ may fail, as in Problem 2.14).

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why this substitution: dividing by y^{α} gives $y^{-\alpha}y' = a\,y^{1-\alpha} + b$, where the left side is, up to a constant factor, the derivative of $y^{1-\alpha}$ and the right side contains $y^{1-\alpha}$ linearly. So the natural unknown is $z = y^{1-\alpha}$.

Change of variables written explicitly: $z = y^{1-\alpha},\ z' = (1-\alpha)\,y^{-\alpha}\,y'$ (chain rule), $\ y = z^{1/(1-\alpha)}$.

PHASE 3: STEP-BY-STEP DERIVATION

Step 1: derive the linear equation

Differentiate $z = y^{1-\alpha}$ with the chain rule and substitute the equation for y':

$$z' = (1-\alpha)\, y^{-\alpha}\, y' = (1-\alpha)\, y^{-\alpha}\, (a\, y + b\, y^{\alpha}) = (1-\alpha)\, \left(a\, y^{1-\alpha} + b\right).$$

Since $y^{1-\alpha} = z$:

$$z' - (1-\alpha)\, a(x)\, z = (1-\alpha)\, b(x).$$

This is linear first order in z, with $p(x) = -(1-\alpha)a(x)$ and $q(x) = (1-\alpha)b(x)$, both continuous.

Step 2: solve it with an integrating factor

Fix $x_0 \in I$ and define $A(x) = \int_{x_0}^{x} a(s)\, ds$. Then $\int p\, dx = -(1-\alpha)A(x)$ and

$$\mu(x) = e^{-(1-\alpha)A(x)}, \qquad \mu(x_0) = 1, \qquad (\mu z)' = (1-\alpha)\, b(x)\, \mu(x).$$

Integrate from x_0 to x (Barrow's rule), with $z_0 = z(x_0) = y_0^{1-\alpha}$:

$$\mu(x) z(x) - z_0 = (1-\alpha) \int_{x_0}^{x} b(s)\, e^{-(1-\alpha)A(s)}\, ds.$$

Multiply by $1/\mu(x) = e^{(1-\alpha)A(x)}$:

$$z(x) = e^{(1-\alpha)A(x)} \left[z_0 + (1-\alpha) \int_{x_0}^{x} b(s)\, e^{-(1-\alpha)A(s)}\, ds \right].$$

Step 3: return to y

Since $y = z^{1/(1-\alpha)}$, wherever the bracket is positive (so that $y > 0$):

$$y(x) = \left\{ e^{(1-\alpha)A(x)} \left[y_0^{1-\alpha} + (1-\alpha) \int_{x_0}^{x} b(s)\, e^{-(1-\alpha)A(s)}\, ds \right] \right\}^{\frac{1}{1-\alpha}}.$$

The solution exists as long as the expression in braces stays positive; if it reaches 0 at a finite point the solution ceases to exist there (the substitution is valid only for $y > 0$).

Step 4: the excluded cases $\alpha = 0$ and $\alpha = 1$

- $\alpha = 1$: the equation is $y' = (a + b)\, y$, already linear and homogeneous. The substitution would give $z = y^0 = 1$, a constant that carries no information (the formula above has the factor $1 - \alpha = 0$, so it degenerates to $z' = 0$). The solution is obtained directly: $y(x) = y_0 \exp\left(\int_{x_0}^{x} (a + b)\, ds\right)$.
- $\alpha = 0$: the equation is $y' = a\, y + b$, already linear (inhomogeneous). The substitution gives $z = y^1 = y$, the identity, and the formula above with $1 - \alpha = 1$ reproduces the variation-of-constants solution $y = e^{A}\left[y_0 + \int_{x_0}^{x} b\, e^{-A}\, ds\right]$. No transformation is needed.

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Check the general solution by differentiation. Put $B(x) = z_0 + (1 - \alpha)\int_{x_0}^{x} b\, e^{-(1-\alpha)A}\, ds$, so $z = e^{(1-\alpha)A} B$ and

$$z' = (1 - \alpha) a\, e^{(1-\alpha)A} B + e^{(1-\alpha)A} (1 - \alpha)\, b\, e^{-(1-\alpha)A} = (1 - \alpha) a\, z + (1 - \alpha) b.$$

This is the linear equation of Step 1, and $z(x_0) = e^0 z_0 = z_0$. Undoing the change of variables is reversible for $y > 0$, so $y = z^{1/(1-\alpha)}$ solves the Bernoulli equation.

Consistency with Problem 2.7: for $a = 1$, $b = x$, $\alpha = -1$ the formula gives $1 - \alpha = 2$, $A = x$, $z' - 2z = 2x$, which is the equation obtained there for $z = y^2$.

Check with a constant-coefficient case (a, b constants, $\alpha = 2$, so $1 - \alpha = -1$): $z' + az = -b$, giving $z = -\frac{b}{a} + \left(z_0 + \frac{b}{a}\right) e^{-a(x - x_0)}$ and $y = 1/z$, which is the logistic-type solution of $y' = ay + by^2$. A sympy substitution of this closed form into $y' = ay + by^2$ (with $a, b > 0$ symbols and $y_0 = 1$) gives residual 0.

RESULT

$z = y^{1-\alpha}$ turns the Bernoulli equation into $z' - (1 - \alpha)az = (1 - \alpha)b$, with solution $z = e^{(1-\alpha)A}\left[z_0 + (1 - \alpha)\int b\, e^{-(1-\alpha)A}\right]$ and $y = z^{1/(1-\alpha)}$. For $\alpha = 1$ and $\alpha = 0$ the equation is already linear and the substitution is trivial or degenerate.